Telecom Customer Churn Prediction Using Machine Learning Models

AI3013 Machine Learning Course Project Final Report

Group	Group 13
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Dataset	IBM Telco Customer Churn Dataset
Task	Binary classification: churn vs. non-churn
Implementation	From scratch with NumPy, Pandas, and Matplotlib
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This report presents a complete from-scratch machine learning workflow for customer churn prediction, including data preprocessing, model theory, implementation, cross-validation, model comparison, threshold analysis, interpretability, and future work.

Abstract

Customer churn prediction is an important real-world machine learning problem in the telecommunications industry. Because retaining an existing customer is usually less expensive than acquiring a new one, accurate early identification of customers likely to leave can support targeted retention campaigns, pricing interventions, and service improvement. This project studies the IBM Telco Customer Churn dataset, which contains demographic, service, contract, and billing attributes for telecom customers together with a binary churn label.

The project implements and compares five classical machine learning models from scratch: Logistic Regression, Cost-Sensitive Logistic Regression, Gaussian Naive Bayes, Hybrid Naive Bayes, and Decision Tree. The implementation uses only basic scientific Python tools such as NumPy, Pandas, and Matplotlib; it does not rely on scikit-learn, TensorFlow, Keras, or other machine learning libraries. A leakage-safe preprocessing pipeline is used, including conversion of TotalCharges to numeric values, removal of missing rows, one-hot encoding, and fold-wise standardization of numerical variables.

Experiments are conducted with stratified 5-fold cross-validation and evaluated using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, log loss, Brier score, training time, and generalization gap. The results show that Cost-Sensitive Logistic Regression gives the best F1-score of about 0.6271 and the strongest balance between recall and precision. Standard Logistic Regression achieves the highest accuracy of about 0.8045 and a very strong ROC-AUC of about 0.8418. Gaussian Naive Bayes achieves the highest recall of about 0.8443, but it produces many false positives and has weaker calibration. Decision Tree provides interpretable nonlinear rules but shows a higher risk of overfitting as depth increases. Overall, the project demonstrates that carefully implemented classical machine learning models can provide useful and interpretable churn prediction performance on structured telecom data.

1. Introduction

1.1 Problem Statement

Customer churn refers to the situation in which a customer stops using a company's services. In the telecom industry, churn is especially important because customers often have multiple alternative providers and can switch services when prices, contracts, service quality, or technical support fail to meet expectations. The objective of this project is to predict whether a customer will churn based on historical customer information. Formally, the task is a supervised binary classification problem: each customer record is represented by a feature vector x, and the target variable y is 1 if the customer churns and 0 otherwise.

The dataset used in this project is suitable for a machine learning course project because it contains a realistic mixture of categorical and numerical variables. The features include contract type, internet service type, payment method, paperless billing, tenure, monthly charges, total charges, and other service-related indicators. These variables are not merely abstract inputs; they are interpretable business factors that can be connected to customer behavior.

1.2 Motivation

The practical value of churn prediction is not only classification accuracy. A telecom company can use churn scores to prioritize retention actions for high-risk customers. For example, customers with month-to-month contracts, short tenure, high monthly charges, or missing technical support may receive targeted offers or service follow-up. Therefore, model evaluation must consider recall and precision as well as accuracy. A model with high accuracy may still miss many churners if the churn class is smaller than the non-churn class.

This project is also motivated by the course requirement to understand machine learning algorithms beyond using library functions. Instead of calling prebuilt classifiers, the models are implemented from scratch. This makes the experiment more transparent: the loss functions, assumptions, optimization procedures, splitting rules, probability estimates, and evaluation metrics can all be inspected directly.

1.3 Main Contributions

- A complete churn prediction workflow is built for a real public dataset, from data cleaning and exploratory analysis to model evaluation and interpretation.
- Five machine learning models are implemented from scratch: Logistic Regression, Cost-Sensitive Logistic Regression, Gaussian Naive Bayes, Hybrid Naive Bayes, and Decision Tree.
- The evaluation uses stratified 5-fold cross-validation and multiple metrics so that class imbalance and business trade-offs are not hidden by accuracy alone.
- The project includes threshold analysis, hyperparameter sensitivity analysis, learning curves, complexity comparison, and feature interpretation.

2. Related Works and Existing Techniques

Customer churn prediction is widely studied because it combines practical business importance with a clear classification structure. Common methods include Logistic Regression, Naive Bayes, Decision Tree, Support Vector Machine, Random Forest, Gradient Boosting, and Neural Networks. In business settings, interpretable models remain valuable because predictions must be converted into retention actions.

Logistic Regression is commonly used as a baseline for churn prediction because it is efficient and interpretable. Its coefficients can be inspected to identify whether a feature increases or decreases churn probability. However, it assumes a linear relationship between the features and the log-odds of the target, so it may underfit if the true decision boundary is strongly nonlinear.

Naive Bayes classifiers are probabilistic models based on Bayes' theorem and the assumption of conditional independence between features given the class label. They are computationally efficient and often perform surprisingly well on high-dimensional data, but the independence assumption is usually unrealistic for telecom churn data. For example, contract type, tenure, payment method, and service subscriptions can be correlated.

Decision Trees are also popular for churn prediction because they can capture nonlinear relationships and produce human-readable decision rules. A tree can split customers according to contract type, tenure, internet service, or payment method in a way that resembles business reasoning. However, unconstrained trees can overfit training data, especially when many categorical dummy variables are available.

Previous churn studies also highlight the importance of class imbalance. In many churn datasets, the number of non-churn customers is larger than the number of churn customers. Therefore, accuracy alone can be misleading. Metrics such as recall, precision, F1-score, ROC-AUC, PR-AUC, and confusion matrices provide a more complete view of model behavior. This project follows that principle and treats threshold selection as part of the modeling problem.

3. Dataset and Exploratory Data Analysis

3.1 Dataset Description

The project uses the IBM Telco Customer Churn dataset from Kaggle [1]. The raw dataset contains 7,043 customer records and 21 original columns. After converting TotalCharges to a numeric variable and removing 11 rows with missing TotalCharges, 7,032 samples remain. The target variable is Churn, which has two classes: Yes and No. After one-hot encoding categorical variables, the design matrix contains 46 input features.

Item	Description
Raw samples	7,043 customers
Samples after cleaning	7,032 customers
Original columns	21 columns including customerID and Churn
Encoded feature count	46 features after one-hot encoding
Target variable	Churn, encoded as Yes = 1 and No = 0
Churn rate	About 26.6% after cleaning

Item	Description
Data source	IBM Telco Customer Churn Dataset, available on Kaggle

Table 1. Dataset summary used in the final experiments.

3.2 Exploratory Findings

The exploratory analysis shows that churn is moderately imbalanced. In the raw data, about 73.5% of customers did not churn and about 26.5% did churn. This imbalance is not extreme, but it is large enough that accuracy alone should not be treated as the main success criterion.

Several customer attributes show clear relationships with churn. Month-to-month contract customers have the highest churn rate, while two-year contract customers have a much lower churn rate. Fiber optic internet service customers show higher churn risk than DSL or customers without internet service. Electronic check payment users also churn more frequently. Short-tenure customers are especially risky: customers with only 0-12 months of tenure have much higher churn than customers with long tenure.

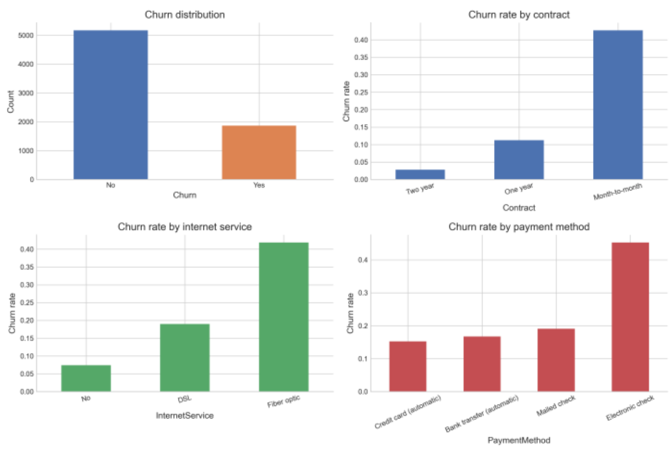

Figure 1. Exploratory overview of churn 1

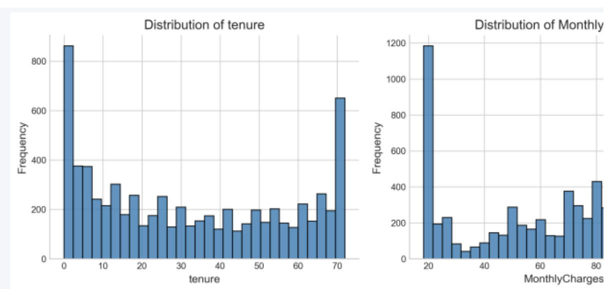

Figure 2. Distributions of numerical feature: tenure, MonthlyCharges, and TotalCharges.

EDA takeaway: Churn is driven by both contract commitment and service experience. Contract type, tenure, internet service, payment method, online security, and technical support are especially informative features for the final models.

4. Methodology

4.1 Preprocessing Pipeline

The preprocessing pipeline is designed to avoid data leakage. A common mistake in machine learning experiments is to standardize the full dataset before cross-validation, allowing information from validation folds to influence training transformations. This project avoids that issue by fitting the mean and standard deviation only on the training fold and then applying the same transformation to the validation fold.

1. Load the raw CSV file and convert TotalCharges from string to numeric values.
2. Remove 11 rows with missing TotalCharges, leaving 7,032 usable samples.
3. Encode the target variable Churn as Yes = 1 and No = 0.
4. Drop customerID because it is an identifier and should not be used as a predictive feature.
5. Apply one-hot encoding to categorical variables without using machine learning libraries.
6. Standardize numerical features tenure, MonthlyCharges, and TotalCharges inside each training fold only.
7. Use stratified 5-fold cross-validation so that each fold preserves the approximate churn/non-churn ratio.

4.2 Logistic Regression

Logistic Regression models the probability of churn with the sigmoid function [3] $p = 1 / (1 + \exp(-(w^T x + b)))$. The model predicts class 1 when the estimated probability is above a selected threshold. The standard threshold is 0.5, but threshold analysis is useful for churn prediction because the business may prefer to capture more churners even at the cost of more false positives.

The loss function is binary cross-entropy: $L = -(1/N) \sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$. L2 regularization is added to control parameter magnitude and reduce overfitting. Parameters are optimized by gradient descent. Because this model is implemented from scratch, the project explicitly computes probabilities, gradients, regularization terms, parameter updates, and final predictions.

4.3 Cost-Sensitive Logistic Regression

Cost-Sensitive Logistic Regression extends standard Logistic Regression by assigning a larger weight to positive churn samples in the loss function. This is useful because the churn class is smaller and because missing an actual churner can be more costly than contacting a customer who would not have churned. In the implementation, the positive-class weight is tuned and compared with the balanced ratio of non-churn to churn customers.

The model keeps the interpretability and efficiency of Logistic Regression while shifting the decision behavior toward higher recall. This makes it a strong candidate for business settings where customer retention campaigns depend on identifying as many high-risk customers as possible.

4.4 Gaussian and Hybrid Naive Bayes

Naive Bayes is based on Bayes' theorem [4][5]: $P(y \mid x)$ is proportional to $P(y) P(x \mid y)$. Under the naive conditional independence assumption, $P(x \mid y)$ is decomposed into a product over features. Gaussian Naive Bayes assumes continuous features follow Gaussian distributions within each class, estimating class-specific means and variances.

The project also implements a Hybrid Naive Bayes variant. Numerical features are modeled with Gaussian distributions, while binary one-hot encoded features are modeled with Bernoulli-style likelihoods. This is more suitable for the Telco dataset because the design matrix contains a mixture of continuous variables and encoded categorical indicators. Both Naive Bayes models are very fast because training mainly requires computing class priors and feature statistics.

4.5 Decision Tree

The Decision Tree classifier recursively partitions the feature space [6]. At each node, the algorithm evaluates candidate features and thresholds and selects the split that reduces impurity the most. This project uses Gini impurity, defined as $Gini(D) = 1 - \sum_k p_k^2$, where p_k is the proportion of class k in node D. The tree stops growing when conditions such as maximum depth, minimum samples per split, or minimum samples per leaf are reached.

Decision Trees can capture nonlinear interactions without requiring explicit feature transformations. For example, a tree may first split on month-to-month contract and then on tenure or internet service. However, deeper trees can fit noise and become unstable. Therefore, max_depth is examined in sensitivity analysis.

4.6 Evaluation Metrics

The final evaluation uses several metrics. Accuracy measures the proportion of correct predictions but can be misleading under class imbalance. Precision measures how many predicted churners actually churned. Recall measures how many actual churners were successfully identified. F1-score balances precision and recall. ROC-AUC evaluates ranking quality over different thresholds, while PR-AUC is especially informative when the positive class is smaller. Log loss and Brier score evaluate probability calibration. Training time and generalization gap are also included to compare computational cost and overfitting.

5. Experimental Study and Result Analysis

5.1 Evaluation Protocol

All models are evaluated with stratified 5-fold cross-validation. For each fold, preprocessing transformations are fitted only on the training subset and applied to the validation subset. The reported cross-validation table is based on validation-fold performance averaged across folds. Out-of-fold predictions are also collected to compute global confusion counts and threshold sweeps.

This protocol is more reliable than a single random train-test split because it reduces dependence on one particular partition and provides a direct estimate of fold-to-fold stability. It also supports a fair comparison across models because each model is evaluated on the same fold structure.

5.2 Overall Model Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Majority baseline	0.7342	0.0000	0.0000	0.0000	-	-
Logistic Regression	0.8045	0.6722	0.5158	0.5837	0.8418	0.6489
Cost-Sensitive Logistic Regression	0.7500	0.5195	0.7908	0.6271	0.8423	0.6494
Gaussian Naive Bayes	0.6950	0.4598	0.8443	0.5954	0.8160	0.5995
Hybrid Naive Bayes	0.7234	0.4876	0.8020	0.6065	0.8195	0.6074
Decision Tree	0.7908	0.6186	0.5554	0.5853	0.8306	0.6278

Table 2. Out-of-fold model performance at threshold 0.5.

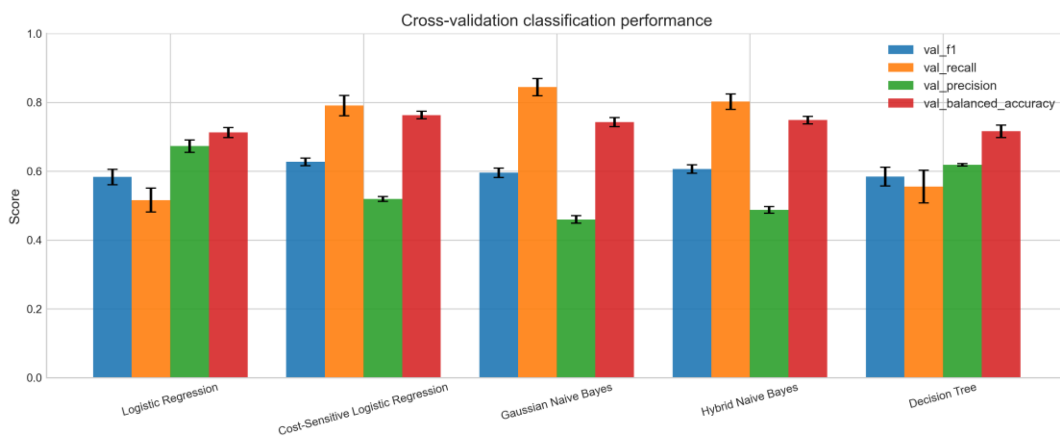

Figure 3. Cross-validation classification performance comparison.

The majority-class baseline appears to have acceptable accuracy only because non-churn customers dominate the dataset, but it has zero recall and zero F1-score for churn detection. The real models therefore need to be judged by recall, F1-score, ROC-AUC, and PR-AUC rather than accuracy alone. The results show that no single model is best on every metric.

Cost-Sensitive Logistic Regression achieves the best F1-score, about 0.6271. Its recall increases to about 0.7908, while precision decreases to about 0.5195. This trade-off is reasonable for churn prediction because the company may prefer to identify more at-risk customers even if some contacted customers would not have churned. Gaussian Naive Bayes has the highest recall, about 0.8443, but its low precision and weaker calibration make it less balanced. Hybrid Naive Bayes improves slightly over Gaussian Naive Bayes in F1-score but still underperforms the cost-sensitive logistic model. Decision Tree performs competitively in accuracy and interpretability but does not beat the best logistic models in F1 or AUC.

5.3 Confusion Matrix Interpretation

Model	TP	TN	FP	FN	Business interpretation
Logistic Regression	964	4693	470	905	Conservative; fewer false alarms but misses many churners.

Model	TP	TN	FP	FN	Business interpretation
Cost-Sensitive Logistic Regression	1478	3796	1367	391	Best balance for retention; captures many more churners.
Gaussian Naive Bayes	1578	3309	1854	291	Very high recall but many false positives.
Hybrid Naive Bayes	1499	3588	1575	370	High recall with fewer false positives than Gaussian NB.
Decision Tree	1038	4523	640	831	More conservative than cost-sensitive methods.

Table 3. Out-of-fold confusion counts at threshold 0.5.

From a business perspective, false negatives are especially important because they are actual churners who are not identified by the model. Standard Logistic Regression has 905 false negatives, while Cost-Sensitive Logistic Regression reduces this number to 391. This explains why the cost-sensitive model is more suitable when the objective is customer retention. However, the cost-sensitive model also produces more false positives, so it may require a larger retention budget.

5.4 Threshold Analysis

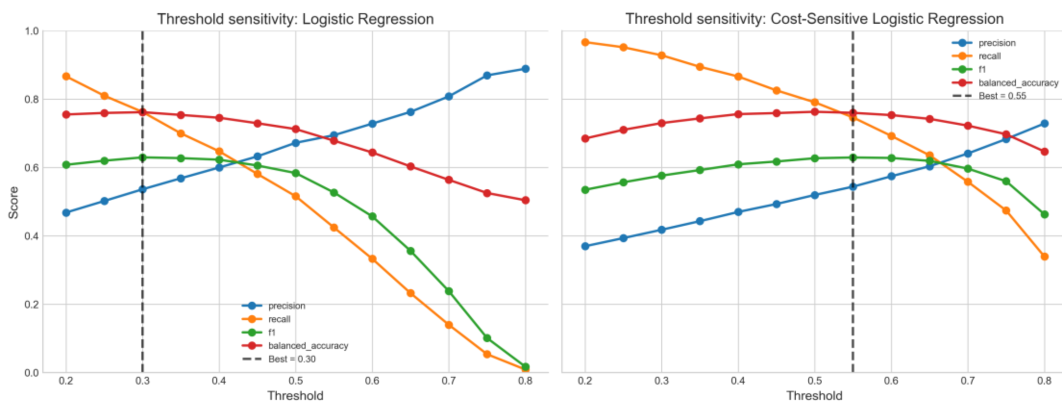

Figure 4. Threshold sensitivity of Logistic Regression and Cost-Sensitive Logistic Regression.

The default threshold of 0.5 is not always optimal for churn prediction. For standard Logistic Regression, lowering the threshold from 0.5 to 0.30 increases recall from about 0.5158 to about 0.7624 and improves F1-score from about 0.5837 to about 0.6296. This shows that a well-ranked model can be adapted to different business objectives simply by changing the decision threshold.

For Cost-Sensitive Logistic Regression, the best F1-score in the tested threshold range occurs around 0.55, with F1 about 0.6292. At threshold 0.5, the model already performs strongly with F1 about 0.6271 and recall about 0.7908. Therefore, if the company wants a simple final deployment rule, Cost-Sensitive Logistic Regression at threshold 0.5 is a practical choice. If the company wants to tune the campaign size more precisely, standard Logistic Regression with a lower threshold is also competitive.

Model / threshold	Accuracy	Precision	Recall	F1	Specificity
Logistic Regression, threshold 0.50	0.8045	0.6722	0.5158	0.5837	0.9090
Logistic Regression, threshold 0.30	0.7615	0.5361	0.7624	0.6296	0.7612
Cost-Sensitive LR, threshold 0.50	0.7500	0.5195	0.7908	0.6271	0.7352
Cost-Sensitive LR, threshold 0.55	0.7662	0.5439	0.7464	0.6292	0.7734

Table 4. Representative threshold trade-offs.

5.5 Generalization and Bias-Variance Analysis

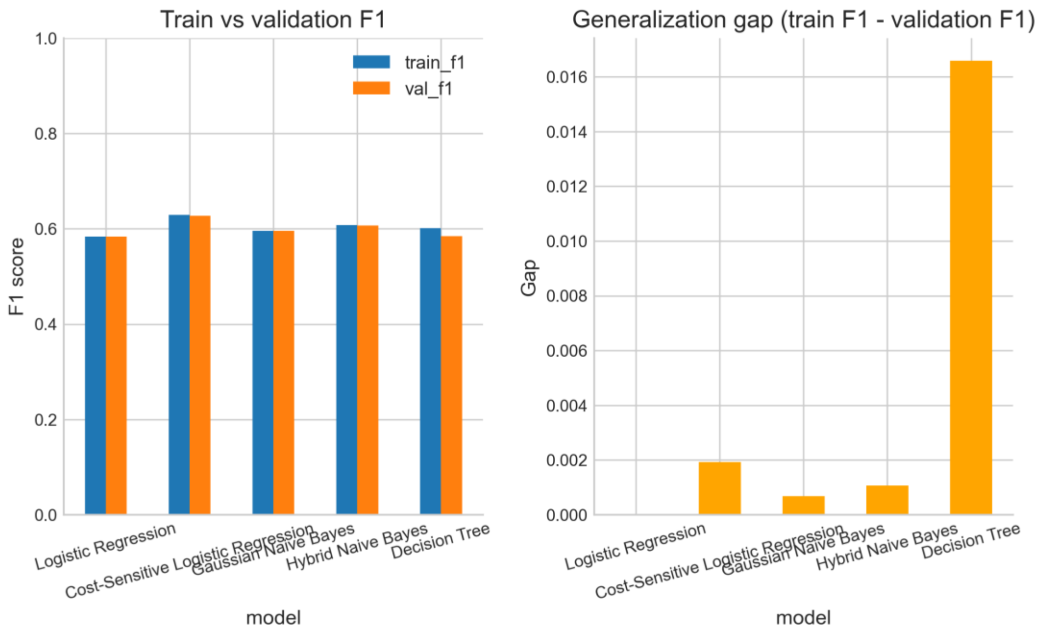

Figure 5. Train-validation F1 comparison and generalization gap.

The train-validation F1 gaps are small for most models. Logistic Regression has almost no generalization gap, and Cost-Sensitive Logistic Regression has a gap of only about 0.0019. Gaussian and Hybrid Naive Bayes also show small gaps because their training procedures are simple and highly constrained. Decision Tree has a larger gap, about 0.0166 at the selected configuration, and the gap increases as the maximum depth becomes larger.

Model	Val F1 mean	Val F1 std	ROC-AUC mean	Log loss	Brier	F1 gap
Logistic Regression	0.5832	0.0223	0.8423	0.4226	0.1370	0.0000
Cost-Sensitive Logistic Regression	0.6269	0.0113	0.8427	0.4971	0.1659	0.0019
Gaussian Naive Bayes	0.5953	0.0139	0.8170	3.0612	0.2893	0.0007
Hybrid Naive Bayes	0.6065	0.0121	0.8199	1.5222	0.2439	0.0011
Decision Tree	0.5844	0.0270	0.8303	0.4385	0.1412	0.0166

Table 5A. Cross-validation stability, calibration, and generalization summary.

The sensitivity analysis confirms the bias-variance trade-off. A shallow decision tree may underfit because it cannot express enough structure, while a very deep tree improves training F1 but reduces validation stability. For Logistic Regression, excessive L2 regularization reduces flexibility and lowers F1. For Cost-Sensitive Logistic Regression, increasing the positive-class weight improves recall but gradually reduces precision and accuracy. These results show that performance depends not only on the model family but also on the objective and hyperparameter setting.

5.6 Computational Complexity

Model	Training complexity	Inference complexity	Main strength	Main weakness
Logistic Regression	O(n d epochs)	O(d)	Interpretable linear baseline	Limited nonlinear expressiveness

Model	Training complexity	Inference complexity	Main strength	Main weakness
Cost-Sensitive Logistic Regression	O(n d epochs)	O(d)	Class-imbalance aware	Requires weight and threshold tuning
Gaussian Naive Bayes	O(n d)	O(d)	Extremely fast probabilistic baseline	Strong Gaussian and independence assumptions
Hybrid Naive Bayes	O(n d)	O(d)	Handles mixed numerical and binary features	Still assumes conditional independence
Decision Tree	O(d n T)	O(depth)	Captures nonlinear relationships	May overfit if too deep

Table 5. Complexity and suitability comparison.

Gaussian and Hybrid Naive Bayes are the fastest models because training mainly computes priors and feature likelihood parameters. Logistic models are slower because they require iterative gradient descent, but their training time remains practical on this dataset. Decision Tree training is more expensive than Naive Bayes because it searches possible splits, but its inference is efficient once the tree is built.

5.7 Feature Interpretation

Interpretability is important because churn prediction should guide business action. The Logistic Regression coefficients and Decision Tree feature importances identify similar churn drivers. Features associated with higher churn include month-to-month contracts, higher monthly charges, fiber optic internet service, electronic check payment, no online security, and no technical support. Features associated with lower churn include longer tenure, two-year contracts, DSL internet service, no paperless billing, online security, and technical support.

Higher churn risk features	Lower churn risk features
Contract: Month-to-month	Longer tenure
Higher MonthlyCharges	Contract: Two year
InternetService: Fiber optic	InternetService: DSL
PaymentMethod: Electronic check	PaperlessBilling: No
OnlineSecurity: No	OnlineSecurity: Yes
TechSupport: No	TechSupport: Yes

Table 6. Interpretable churn drivers from model analysis.

These results are consistent with business intuition. Customers with short commitments and limited support are more likely to leave, while long-tenure customers and customers with longer contracts are more stable. The feature interpretation also suggests retention strategies: improve technical support, monitor short-tenure customers, and design offers for month-to-month customers with high monthly charges.

6. Discussion

The experimental results show that model selection should depend on the business objective. Standard Logistic Regression is attractive for accuracy and stable ranking, while Cost-Sensitive Logistic Regression is stronger for retention targeting because it identifies many more actual churners and achieves the best F1-score.

Gaussian Naive Bayes demonstrates the value and limitation of strong assumptions. It is very fast and detects many churners, but it makes too many false positive predictions and performs poorly on probability calibration. This is likely because the Telco dataset contains correlated service and contract variables that violate the conditional independence assumption. The Hybrid Naive Bayes model partially adapts the likelihood to mixed feature types, improving F1 over Gaussian Naive Bayes, but the independence assumption still limits performance.

The Decision Tree model is useful for explanation because it identifies a small number of important decision variables, especially month-to-month contract, tenure, and fiber optic internet service. However, the tree is more sensitive to depth and sample variation. Its performance is competitive but less stable than the best logistic model. In a real business application, a tree could be used as a supplementary interpretation tool even if a logistic model is selected for prediction.

A key lesson from the project is that the threshold is part of the model design. A model's probability ranking and its binary classification behavior are not the same. By adjusting the threshold, a company can choose how many customers to target. This is especially useful in churn prediction because the cost of contacting customers and the cost of losing customers are not equal.

7. Limitations and Future Work

Although the project satisfies the course requirement of implementing machine learning models from scratch and conducting a complete experimental analysis, several limitations remain. First, the study uses one dataset only. The conclusions may depend on the specific feature distribution of the IBM Telco dataset. Testing the same implementation on another churn dataset would help evaluate generalizability.

Second, the project does not include advanced ensemble models such as Random Forest or Gradient Boosting because the course emphasizes classical machine learning implementation from scratch. Such methods may improve predictive performance, but they would also require more complex implementation and interpretation. Third, the current threshold analysis is based on F1-score and general business reasoning. A real telecom company would ideally provide explicit costs for false positives, false negatives, retention offers, and customer lifetime value. These values could be used to optimize expected profit rather than classification metrics alone.

Future work can extend the project in several directions. The Decision Tree split strategy could be improved with more efficient threshold search and pruning. Probability calibration could be applied to Naive Bayes and Decision Tree outputs. Cost-sensitive learning could be expanded from Logistic Regression to other models. Finally, the project could include a simple retention policy simulation that estimates how many churners would be captured under different campaign budgets.

8. Conclusion

This project developed a complete machine learning workflow for telecom customer churn prediction using models implemented from scratch. The final dataset contains 7,032 cleaned samples and 46 encoded features. The project compared Logistic Regression, Cost-Sensitive Logistic Regression, Gaussian Naive Bayes, Hybrid Naive Bayes, and Decision Tree under a consistent stratified 5-fold cross-validation protocol.

The best overall F1-score is achieved by Cost-Sensitive Logistic Regression, which balances recall and precision better than the other models. Standard Logistic Regression achieves the highest accuracy and strong ROC-AUC, making it a strong baseline and a good option when conservative predictions are preferred. Gaussian Naive Bayes achieves the highest recall but has many false positives, while Decision Tree provides useful interpretation but shows higher overfitting risk.

The final recommendation is Cost-Sensitive Logistic Regression for retention-focused prediction. If the company has a smaller retention budget or wants fewer false positives, standard Logistic Regression with a carefully selected threshold is also strong. The project also shows that transparent from-scratch implementations can reveal both algorithm behavior and customer churn patterns.

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Contribution of Each Team Member

Team member	Contribution
Cao Yiqian	Code implementation, experimental analysis, and PPT preparation.
Gao Ruiyang	Final report writing and project documentation.
Wang Runa	Preliminary preparation, topic/background support, and early-stage project organization.
Huang Yueqing	Preliminary preparation, model/background research, and early-stage presentation support.

Table 7. Current team contribution summary.